

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Integrals

1.1 Medium Integral

Lambert W Function Substitution

Evaluate the definite integral:

$$\int_0^e \frac{\ln\left(\frac{1+W(x)}{1-W(x)}\right)}{x(1+W(x))} dx$$

where $W(x)e^{W(x)} = x$.

Solution: We use the substitution $u = W(x)$. From the defining equation of the Lambert W function, we have $x = ue^u$. Differentiating both sides with respect to u , we get:

$$dx = (e^u + ue^u) du = e^u(1 + u) du$$

The integration bounds transform as follows: When $x = 0$, $u = W(0) = 0$. When $x = e$, $u = W(e) = 1$. Substitute these into the integral:

$$I = \int_0^1 \frac{\ln\left(\frac{1+u}{1-u}\right)}{ue^u(1+u)} e^u(1+u) du$$

The terms $e^u(1+u)$ cancel out beautifully:

$$I = \int_0^1 \frac{\ln(1+u) - \ln(1-u)}{u} du = \int_0^1 \frac{\ln(1+u)}{u} du - \int_0^1 \frac{\ln(1-u)}{u} du$$

These are well-known integrals related to the dilogarithm function $\text{Li}_2(z) = -\int_0^z \frac{\ln(1-t)}{t} dt$. We have:

$$\int_0^1 \frac{\ln(1-u)}{u} du = -\text{Li}_2(1) = -\frac{\pi^2}{6}$$

$$\int_0^1 \frac{\ln(1+u)}{u} du = -\text{Li}_2(-1) = \frac{\pi^2}{12}$$

Subtracting the two results yields:

$$I = \frac{\pi^2}{12} - \left(-\frac{\pi^2}{6}\right) = \frac{\pi^2}{12} + \frac{2\pi^2}{12} = \frac{3\pi^2}{12} = \frac{\pi^2}{4}$$

1.2 Hard Integral

Logarithmic and Rational Function

Evaluate the definite integral:

$$\int_0^{+\infty} \frac{\ln(1+x^2)}{(1+x^2)^2} dx$$

Solution: We employ the trigonometric substitution $x = \tan(\theta)$, which implies $dx = \sec^2(\theta) d\theta$. The limits of integration change from $[0, +\infty)$ to $[0, \pi/2)$. Substituting this into the integral gives:

$$I = \int_0^{\pi/2} \frac{\ln(1+\tan^2(\theta))}{(1+\tan^2(\theta))^2} \sec^2(\theta) d\theta$$

Using the Pythagorean identity $1 + \tan^2(\theta) = \sec^2(\theta)$:

$$I = \int_0^{\pi/2} \frac{\ln(\sec^2(\theta))}{\sec^4(\theta)} \sec^2(\theta) d\theta = \int_0^{\pi/2} \frac{2 \ln(\sec(\theta))}{\sec^2(\theta)} d\theta$$

Since $\ln(\sec(\theta)) = -\ln(\cos(\theta))$ and $\frac{1}{\sec^2(\theta)} = \cos^2(\theta)$, we can rewrite this as:

$$I = -2 \int_0^{\pi/2} \cos^2(\theta) \ln(\cos(\theta)) d\theta$$

Apply the double-angle identity $\cos^2(\theta) = \frac{1+\cos(2\theta)}{2}$:

$$I = -2 \int_0^{\pi/2} \frac{1+\cos(2\theta)}{2} \ln(\cos(\theta)) d\theta = - \int_0^{\pi/2} \ln(\cos(\theta)) d\theta - \int_0^{\pi/2} \cos(2\theta) \ln(\cos(\theta)) d\theta$$

The first integral is a classic result:

$$\int_0^{\pi/2} \ln(\cos(\theta)) d\theta = -\frac{\pi}{2} \ln(2)$$

For the second integral, we use integration by parts with $u = \ln(\cos(\theta))$ and $dv = \cos(2\theta) d\theta$. Then $du = -\tan(\theta) d\theta$ and $v = \frac{1}{2} \sin(2\theta) = \sin(\theta) \cos(\theta)$.

$$\int_0^{\pi/2} \cos(2\theta) \ln(\cos(\theta)) d\theta = [\ln(\cos(\theta)) \sin(\theta) \cos(\theta)]_0^{\pi/2} - \int_0^{\pi/2} \sin(\theta) \cos(\theta) (-\tan(\theta)) d\theta$$

The boundary term evaluates to 0 at both limits. The remaining integral is:

$$\int_0^{\pi/2} \sin^2(\theta) d\theta = \frac{\pi}{4}$$

Bringing it all together:

$$I = - \left(-\frac{\pi}{2} \ln(2) \right) - \frac{\pi}{4} = \frac{\pi}{2} \ln(2) - \frac{\pi}{4}$$

Differentials

2.1 Medium Differential

Implicit Variable Substitution ODE

Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be the unique solution of:

$$\varphi'(x) = e^{x-\varphi(x)} + \varphi(x) - x$$

satisfying $\varphi(0) = 0$. Determine exactly $\varphi(\ln(2))$.

Solution: Let us define a new function $y(x) = \varphi(x) - x$. Differentiating this with respect to x yields $y'(x) = \varphi'(x) - 1$. We can rewrite the original differential equation in terms of y :

$$y'(x) + 1 = e^{-y(x)} + y(x) \implies y'(x) = e^{-y(x)} + y(x) - 1$$

The initial condition is given by $\varphi(0) = 0$, which translates to $y(0) = \varphi(0) - 0 = 0$. Notice that if we substitute $y(x) = 0$ into the differential equation, we get:

$$0 = e^0 + 0 - 1 \implies 0 = 1 - 1 \implies 0 = 0$$

Thus, the constant function $y(x) = 0$ is a valid solution to the differential equation. By the Picard-Lindelöf theorem (since the right-hand side is continuously differentiable), this solution is unique for the given initial value problem. Therefore, $y(x) = 0$ for all $x \in \mathbb{R}$. Substituting back our definition of $y(x)$:

$$\varphi(x) - x = 0 \implies \varphi(x) = x$$

Finally, we evaluate this function at $x = \ln(2)$:

$$\varphi(\ln(2)) = \ln(2)$$

2.2 Hard Differential

Non-linear Second-Order Initial Value Problem

Consider the non-linear second-order initial value problem:

$$yy'' - (y')^2 = y^2 \ln(y)$$

Given the initial conditions $y(0) = e$ and $y'(0) = e$, find the exact value of $y(1)$.

Solution: We start by dividing the entire differential equation by y^2 :

$$\frac{yy'' - (y')^2}{y^2} = \ln(y)$$

Notice that the left-hand side is precisely the derivative of the quotient $\frac{y'}{y}$:

$$\frac{d}{dx} \left(\frac{y'}{y} \right) = \frac{d^2}{dx^2} (\ln(y))$$

Let $u(x) = \ln(y(x))$. The equation simplifies remarkably to a linear second-order ODE:

$$u'' = u$$

The general solution to this linear ODE is:

$$u(x) = c_1 e^x + c_2 e^{-x}$$

We now translate the initial conditions for y into initial conditions for u . Since $y(0) = e$, we have $u(0) = \ln(e) = 1$. Since $y'(x) = u'(x)e^{u(x)}$, evaluating at $x = 0$ gives $y'(0) = u'(0)e^{u(0)}$. Substitute the known values: $e = u'(0)e^1 \implies u'(0) = 1$. Apply these initial conditions to our general solution: 1. $u(0) = c_1 e^0 + c_2 e^0 = c_1 + c_2 = 1$ 2. $u'(x) = c_1 e^x - c_2 e^{-x} \implies u'(0) = c_1 - c_2 = 1$ Adding the two equations yields $2c_1 = 2 \implies c_1 = 1$. Subtracting the two equations yields $2c_2 = 0 \implies c_2 = 0$. The specific solution for $u(x)$ is therefore $u(x) = e^x$. Reverting back to $y(x)$ via $y(x) = e^{u(x)}$:

$$y(x) = e^{e^x}$$

To find $y(1)$, we substitute $x = 1$:

$$y(1) = e^{e^1} = e^e$$

Derivatives

3.1 Medium Derivative

Derivative of a Function to a Variable Power

Let $f(x) = (\ln(x))^x$. Find $f'(e)$.

Solution: To find the derivative of a function where both the base and exponent depend on x , we use logarithmic differentiation. Let $y = (\ln(x))^x$. Take the natural logarithm of both sides:

$$\ln(y) = \ln((\ln(x))^x) = x \ln(\ln(x))$$

Now, differentiate both sides with respect to x using the product rule on the right-hand side:

$$\begin{aligned} \frac{1}{y} y' &= \frac{d}{dx} [x \ln(\ln(x))] \\ \frac{1}{y} y' &= (1) \ln(\ln(x)) + x \left(\frac{1}{\ln(x)} \cdot \frac{1}{x} \right) \end{aligned}$$

Simplify the right side:

$$\frac{1}{y} y' = \ln(\ln(x)) + \frac{1}{\ln(x)}$$

Multiply both sides by y to solve for y' :

$$f'(x) = y' = (\ln(x))^x \left[\ln(\ln(x)) + \frac{1}{\ln(x)} \right]$$

Now we evaluate the derivative at $x = e$:

$$f'(e) = (\ln(e))^e \left[\ln(\ln(e)) + \frac{1}{\ln(e)} \right]$$

Since $\ln(e) = 1$:

$$\begin{aligned} f'(e) &= (1)^e \left[\ln(1) + \frac{1}{1} \right] \\ f'(e) &= 1 \cdot (0 + 1) = 1 \end{aligned}$$

3.2 Hard Derivative

Golden Ratio Evaluation

Let the function $f : (1, 2) \rightarrow \mathbb{R}$ be defined by:

$$f(x) = x^x \cdot \operatorname{Re} \left[\int_0^{\pi/2} \ln(\lfloor x \rfloor + i\{x\} \tan(\theta)) d\theta \right]$$

Find $f'(\phi)$ where $\phi = \frac{1+\sqrt{5}}{2}$.

Solution: Let $I = \int_0^{\pi/2} \ln(a + ib \tan(\theta)) d\theta$, where $a = \lfloor x \rfloor$ and $b = \{x\}$. For $x \in (1, 2)$, $a = 1 > 0$ and $b = x - 1 > 0$. The real part of this integral is equivalent to:

$$\operatorname{Re}(I) = \frac{1}{2} \int_0^{\pi/2} \ln(a^2 + b^2 \tan^2(\theta)) d\theta = \frac{1}{2} \int_0^{\pi/2} \ln(a^2 \cos^2(\theta) + b^2 \sin^2(\theta)) d\theta - \int_0^{\pi/2} \ln(\cos(\theta)) d\theta$$

Using the known definite integrals $\int_0^{\pi/2} \ln(a^2 \cos^2(\theta) + b^2 \sin^2(\theta)) d\theta = \pi \ln\left(\frac{a+b}{2}\right)$ and $\int_0^{\pi/2} \ln(\cos(\theta)) d\theta = -\frac{\pi}{2} \ln(2)$:

$$\operatorname{Re}(I) = \frac{\pi}{2} \ln\left(\frac{a+b}{2}\right) + \frac{\pi}{2} \ln(2) = \frac{\pi}{2} \ln(a+b)$$

Since a is the integer part of x and b is the fractional part, $a + b = x$. Thus, the integral evaluates to $\frac{\pi}{2} \ln(x)$. The function simplifies to:

$$f(x) = \frac{\pi}{2} x^x \ln(x)$$

Differentiating using the product rule, and knowing $\frac{d}{dx}(x^x) = x^x(\ln(x) + 1)$:

$$f'(x) = \frac{\pi}{2} \left[x^x(\ln(x) + 1) \ln(x) + x^x \cdot \frac{1}{x} \right] = \frac{\pi}{2} x^x \left(\ln^2(x) + \ln(x) + \frac{1}{x} \right)$$

Evaluate at $x = \phi$. Note that ϕ is the golden ratio, satisfying $\phi^2 - \phi - 1 = 0$, which gives $\frac{1}{\phi} = \phi - 1$.

$$f'(\phi) = \frac{\pi}{2} \phi^\phi \left(\ln^2(\phi) + \ln(\phi) + \phi - 1 \right)$$

Physics & Engineering

4.1 Electric Flux

Semi-infinite Thread Flux

A semi-infinite, uniformly charged thread is oriented along the central axis of a circular area of radius $R = 0.50$ m. One end of the thread is at the center of the circle. The thread has a linear charge density $\lambda = 2.0 \mu\text{C m}^{-1}$. Calculate the magnitude of the electric flux Φ across the circle area in kV m. Take $\epsilon_0 = \frac{1}{36\pi \times 10^9} \text{ F m}^{-1}$.

Solution: Consider the electric field dE produced by an infinitesimal segment dz of the thread located at distance z from the origin on the z -axis. The z -component of the electric field at a point on the circular disk (distance r from the center) is:

$$dE_z = -\frac{1}{4\pi\epsilon_0} \frac{\lambda dz}{r^2 + z^2} \cos(\alpha) = -\frac{\lambda z dz}{4\pi\epsilon_0 (r^2 + z^2)^{3/2}}$$

Integrating from $z = 0$ to $z = \infty$:

$$E_z = -\frac{\lambda}{4\pi\epsilon_0} \int_0^\infty \frac{z}{(r^2 + z^2)^{3/2}} dz = -\frac{\lambda}{4\pi\epsilon_0} \left[\frac{-1}{\sqrt{r^2 + z^2}} \right]_0^\infty = -\frac{\lambda}{4\pi\epsilon_0 r}$$

The total flux through the circular area is obtained by integrating over the disk:

$$\Phi = \int_0^R |E_z| 2\pi r dr = \int_0^R \frac{\lambda}{4\pi\epsilon_0 r} 2\pi r dr = \frac{\lambda}{2\epsilon_0} \int_0^R dr = \frac{\lambda R}{2\epsilon_0}$$

Plugging in the given values, with $\lambda = 2.0 \times 10^{-6} \text{ C m}^{-1}$ and $R = 0.50$ m:

$$\Phi = \frac{2.0 \times 10^{-6} \times 0.50}{2 \times \frac{1}{36\pi \times 10^9}} = \frac{1.0 \times 10^{-6}}{\frac{2}{36\pi \times 10^9}} = (1.0 \times 10^{-6}) \times (18\pi \times 10^9) = 18\pi \times 10^3 \text{ V m}$$

Converting to kV m, the magnitude of the electric flux is:

$$\Phi = 18\pi \text{ kV m}$$

4.2 Quantum Mechanics

Infinite Potential Well Probability

A particle is confined in a one-dimensional infinite potential well of width L , with normalized initial wavefunction $\psi(x, 0) = \sqrt{\frac{8}{3L}} \sin^2\left(\frac{\pi x}{L}\right)$ for $0 < x < L$, and 0 elsewhere. Find the probability of measuring the particle's energy to be $E = \frac{2\pi^2 \hbar^2}{mL^2}$.

Solution: The energy levels of a particle in a one-dimensional infinite potential well are given by $E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}$. Setting the given energy equal to this general formula:

$$\frac{2\pi^2\hbar^2}{mL^2} = \frac{n^2\pi^2\hbar^2}{2mL^2} \implies 4 = n^2 \implies n = 2$$

We seek the probability of measuring the state corresponding to the first excited state, $n = 2$. The normalized eigenfunction for this state is $\phi_2(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{2\pi x}{L}\right)$. The probability is given by the squared magnitude of the inner product of $\phi_2(x)$ and the initial state $\psi(x, 0)$:

$$P_2 = |\langle \phi_2 | \psi(x, 0) \rangle|^2 = \left| \int_0^L \sqrt{\frac{2}{L}} \sin\left(\frac{2\pi x}{L}\right) \sqrt{\frac{8}{3L}} \sin^2\left(\frac{\pi x}{L}\right) dx \right|^2$$

Let us evaluate the integral. Let $u = \frac{\pi x}{L}$, so $dx = \frac{L}{\pi} du$.

$$\langle \phi_2 | \psi(x, 0) \rangle \propto \int_0^\pi \sin(2u) \sin^2(u) du$$

Using the double-angle formula $\sin(2u) = 2 \sin(u) \cos(u)$, the integral becomes:

$$\int_0^\pi 2 \sin^3(u) \cos(u) du$$

This evaluates to $\left[\frac{1}{2} \sin^4(u) \right]_0^\pi = \frac{1}{2}(0 - 0) = 0$. Since the inner product is exactly zero, the probability is:

$$P_2 = 0$$

4.3 Mechanics

Bouncing Ball Kinematics

A ball is thrown vertically downward from the top of a 523 m high building with an initial speed of 50 m s^{-1} . After striking the ground for the first time, it rebounds vertically upward with a speed equal to $8/9$ of the speed with which it struck the ground. After every subsequent bounce, the maximum height reached by the ball is 5 m less than the maximum height reached after the previous bounce. How many times does the ball hit the ground? Take $g = 10 \text{ m s}^{-2}$.

Solution: First, we find the velocity of the ball just before it hits the ground for the first time using the kinematic equation $v_1^2 = u^2 + 2gH$:

$$v_1^2 = 50^2 + 2(10)(523) = 2500 + 10460 = 12960 \implies v_1 = \sqrt{12960} = 36\sqrt{10} \text{ m s}^{-1}$$

The rebound speed after the first bounce is $u_1 = \frac{8}{9}v_1$:

$$u_1 = \frac{8}{9}(36\sqrt{10}) = 32\sqrt{10} \text{ m s}^{-1}$$

The maximum height reached after the first bounce is $h_1 = \frac{u_1^2}{2g}$:

$$h_1 = \frac{(32\sqrt{10})^2}{20} = \frac{1024 \times 10}{20} = 512 \text{ m}$$

The maximum height decreases by 5 m after each subsequent bounce. The height after the n -th bounce is given by the arithmetic progression:

$$h_n = h_1 - 5(n - 1) = 512 - 5(n - 1)$$

For the ball to hit the ground after a bounce, it must reach a height greater than zero. We find the largest n such that $h_n > 0$:

$$512 - 5(n - 1) > 0 \implies 5(n - 1) < 512 \implies n - 1 < 102.4 \implies n < 103.4$$

The ball achieves a positive maximum height up to the 103rd bounce:

$$h_{103} = 512 - 5(102) = 2 \text{ m}$$

After reaching 2 m, it falls and hits the ground for the 104th time. Following this 104th impact, the formula yields a negative height, indicating the ball ceases to bounce upwards. Thus, it hits the ground exactly 104 times.